Power Plant Boiler & Turbine, BFP Operations and Preventive Maintenance

These sample Standard Operating Procedures (SOPs) outline the standard steps for safely starting, running, and maintaining thermal power plant components. While specific parameters vary by design, these frameworks represent industry-standard best practices. [1, 2, 3, 4, 5]

1. Boiler Operations & Maintenance

Startup Procedure (Cold/Warm)

1. **System Walkdown:** Ensure all Maintenance Permits to Work (PTW) are cleared and closed. Visually inspect the boiler for loose debris.
2. **Water & Vents:** Check that the deaerator levels are normal. Open the steam drum start-up air vents and superheater header vents.
3. **Draft System:** Start the Forced Draft (FD) and Induced Draft (ID) fans to purge the furnace of combustible gases for 5-10 minutes.
4. **Light-Up:** Start the main fuel burners using Light Diesel Oil (LDO) or Heavy Fuel Oil (HFO). Monitor the flame established via the burner management system.
5. **Ramping Up:** Gradually increase fuel firing rates to manage the thermal stress on pressure parts. Close the drum vents once steam pressure reaches 1.0 kg/cm^2. [6, 7, 8, 9, 10]

Boiler Preventive Maintenance (PM) Checklist

- **Daily:** Inspect the water level gauge glasses and blow down the boiler to remove suspended solids. Record all flue gas temperatures, draft pressures, and feed water quality parameters.
- **Weekly:** Inspect FD/ID fan foundation bolts and drive belts. Test the operation of the safety valves and soot blowers.
- **Monthly:** Calibrate all pressure, temperature, and draft transmitters. Check the burner igniters and flame scanners for carbon buildup.
- **Annual:** Conduct internal visual inspections of the steam drum for scale, carryover, or corrosion. Hydrostatically test tubes for leaks. [1, 9, 11, 12, 13, 14]

2. Turbine Operations & Maintenance

Startup Procedure

1. **Pre-Start Checks:** Ensure all PTWs are canceled. Check that the lube oil pumps and Auxiliary Cooling Water (ACW) systems are operating normally.
2. **Turning Gear:** Engage the turbine turning gear (rotor barring) to ensure even thermal distribution as temperatures rise.
3. **Condenser Vacuum:** Start the Condensate Extraction Pump (CEP) and the vacuum pumps (ejectors) to pull the necessary vacuum in the condenser.
4. **Warming and Rolling:** Open the Main Steam Stop Valve (MSSV) bypass to warm the steam lines. Accelerate the turbine to rated speed (e.g., 3000 rpm for 50 Hz units) while monitoring for critical vibration spikes.
5. **Synchronization:** Match the turbine generator's voltage, frequency, and phase with the electrical grid. Close the generator breaker and load the turbine gradually. [7, 8, 13, 15, 16]

Turbine Preventive Maintenance (PM) Checklist

- **Daily:** Record bearing metal and oil temperatures. Monitor turbine vibration, eccentricity, and differential expansion logs.
- **Weekly:** Check the lube oil quality for moisture or particulate contamination. Test the overspeed trip mechanism (if plant configurations allow).
- **Monthly:** Test the turbine Emergency Trip valves and observe control valves for smooth, unhindered operation.
- **Annual:** Perform non-destructive testing (NDT) on rotor blades and casing studs. Check alignment between the turbine and generator using laser alignment tools. [1, 7, 13]

3. Boiler Feed Pump (BFP) Operations & Maintenance

Startup Procedure

1. **Lining Up:** Ensure the pump maintenance permit is cleared. Open the suction valve and ensure the Deaerator level is at normal operating height.
2. **Cooling & Lube Oil:** Turn on the auxiliary lube oil pump and ensure the oil pressure is healthy (e.g., $>1.8\text{ kg/cm}^2$). Start the mechanical seal cooling water and motor jacket cooling.
3. **Venting & Positioning:** Open all pump air vents to remove trapped air. Open the minimum recirculation valve and keep the main discharge valve closed.
4. **Starting:** Start the BFP motor. Wait for the pump to reach full speed, then slowly crack open the discharge valve from the control room DCS until the line pressure normalizes.

5. **Regulating:** Ensure the pump's minimum recirculation valve closes automatically upon reaching a set-point flow. Check that balancing leak-off and seal water pressures are healthy. [7, 17, 18, 19, 20]

BFP Preventive Maintenance (PM) Checklist

- **Daily:** Monitor suction and discharge pressures, bearing temperatures (typically keep below 85°C), and motor winding temperatures. Inspect for any seal water or oil leaks.
- **Weekly:** Check the differential pressure (DP) across the suction strainer. Clean or replace the filters if the DP exceeds manufacturer thresholds.
- **Monthly:** Check vibration levels (velocity must be within safe limits, e.g., $<18\text{ mm/s}$). Check the coupling alignment and clearance.
- **Annual:** Overhaul the mechanical seals and inspect pump impellers for cavitation damage. Measure and record rotor axial play and thrust bearing wear. [7, 17, 21, 22, 23, 24, 25]

For a deeper dive into your specific facility's requirements, you can consult operational guidelines from the Electric Power Research Institute (EPRI) or detailed Fossil Consulting Services (FCS) standards.